

Simulation run checklist

Generated by `analysis/sim_case_list.py`. 380 runs in total. Each entry is one (case, axis, ramp rate); both tip directions are run under it, so one tick covers two excitations. Rates in N m/s.

- ☐ **1. Rotor ablation, ideal rotor** (56 runs) — run this first
 - ☐ **2. Main sweep, identified rotor** (324 runs)
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1. Rotor ablation — ideal rotor (R0)

Compare against the same cases in the main sweep, which uses the identified second-order-plus-delay rotor (R3). Plot the offset error against the ramp rate: R0 should be flat and R3 should not.

S2 (R0) — (0, 10) mm

roll ☐ 0.10 ☐ 0.20 ☐ 0.30 ☐ 0.45 ☐ 0.65 ☐ 0.90 ☐ 1.20
pitch ☐ 0.10 ☐ 0.20 ☐ 0.30 ☐ 0.45 ☐ 0.65 ☐ 0.90 ☐ 1.20

S8 (R0) — (25, 25) mm

roll ☐ 0.10 ☐ 0.20 ☐ 0.30 ☐ 0.45 ☐ 0.65 ☐ 0.90 ☐ 1.20
pitch ☐ 0.10 ☐ 0.20 ☐ 0.30 ☐ 0.45 ☐ 0.65 ☐ 0.90 ☐ 1.20

2. Main sweep — identified rotor (R3)

S1 — small offset, (-6, 0) mm, 3.066 kg

predicted M_{crit} : roll -0.99 / +0.99, pitch -1.08 / +1.44
roll ☐ 0.10 ☐ 0.20 ☐ 0.30 ☐ 0.45 ☐ 0.65 ☐ 0.90 ☐ 1.20
pitch ☐ 0.10 ☐ 0.20 ☐ 0.30 ☐ 0.45 ☐ 0.65 ☐ 0.90 ☐ 1.20
☐ identified offset within 1 mm of truth

S2 — small offset, (0, 10) mm, 3.066 kg

predicted M_{crit} : roll -0.69 / +1.29, pitch -1.26 / +1.26
roll ☐ 0.10 ☐ 0.20 ☐ 0.30 ☐ 0.45 ☐ 0.65 ☐ 0.90 ☐ 1.20
pitch ☐ 0.10 ☐ 0.20 ☐ 0.30 ☐ 0.45 ☐ 0.65 ☐ 0.90 ☐ 1.20
☐ identified offset within 1 mm of truth

S3 — small offset, (10, -5) mm, 3.066 kg

predicted M_{crit} : roll -1.14 / +0.84, pitch -1.56 / +0.96
roll ☐ 0.10 ☐ 0.20 ☐ 0.30 ☐ 0.45 ☐ 0.65 ☐ 0.90 ☐ 1.20
pitch ☐ 0.10 ☐ 0.20 ☐ 0.30 ☐ 0.45 ☐ 0.65 ☐ 0.90 ☐ 1.20
☐ identified offset within 1 mm of truth

S4 — box corner, (20, 20) mm, 3.066 kg

predicted M_{crit} : roll -0.39 / +1.59, pitch -1.86 / +0.66
roll ☐ 0.10 ☐ 0.20 ☐ 0.30 ☐ 0.45 ☐ 0.65 ☐ 0.90 ☐ 1.20
pitch ☐ 0.10 ☐ 0.20 ☐ 0.30 ☐ 0.45 ☐ 0.65 ☐ 0.90 ☐ 1.20
☐ identified offset within 1 mm of truth

S5 — box corner, (20, -20) mm, 3.066 kg

predicted M_{crit} : roll -1.59 / +0.39, pitch -1.86 / +0.66
roll ☐ 0.10 ☐ 0.20 ☐ 0.30 ☐ 0.45 ☐ 0.65 ☐ 0.90 ☐ 1.20
pitch ☐ 0.10 ☐ 0.20 ☐ 0.30 ☐ 0.45 ☐ 0.65 ☐ 0.90 ☐ 1.20
☐ identified offset within 1 mm of truth

S6 — box corner, (-20, 20) mm, 3.066 kg

predicted M_{crit} : roll -0.39 / +1.59, pitch -0.66 / +1.86
roll ☐ 0.10 ☐ 0.20 ☐ 0.30 ☐ 0.45 ☐ 0.65 ☐ 0.90 ☐ 1.20
pitch ☐ 0.10 ☐ 0.20 ☐ 0.30 ☐ 0.45 ☐ 0.65 ☐ 0.90 ☐ 1.20
☐ identified offset within 1 mm of truth

S7 — box corner, (-20, -20) mm, 3.066 kg

predicted M_{crit} : roll -1.59 / +0.39, pitch -0.66 / +1.86

roll ☐ 0.10 ☐ 0.20 ☐ 0.30 ☐ 0.45 ☐ 0.65 ☐ 0.90 ☐ 1.20

pitch ☐ 0.10 ☐ 0.20 ☐ 0.30 ☐ 0.45 ☐ 0.65 ☐ 0.90 ☐ 1.20

☐ identified offset within 1 mm of truth

S8 — roll boundary, (25, 25) mm, 3.066 kg

predicted M_{crit} : roll -0.24 / +1.74, pitch -2.02 / +0.51

roll ☐ 0.10 ☐ 0.20 ☐ 0.30 ☐ 0.45 ☐ 0.65 ☐ 0.90 ☐ 1.20

pitch ☐ 0.10 ☐ 0.20 ☐ 0.30 ☐ 0.45 ☐ 0.65 ☐ 0.90 ☐ 1.20

☐ identified offset within 1 mm of truth

S9 — roll boundary, (32, 32) mm, 3.066 kg

predicted M_{crit} : roll -0.03 / +1.96, pitch -2.23 / +0.30

roll ☐ 0.10 ☐ 0.20 ☐ 0.30 ☐ 0.45 ☐ 0.65 ☐ 0.90 ☐ 1.20

pitch ☐ 0.10 ☐ 0.20 ☐ 0.30 ☐ 0.45 ☐ 0.65 ☐ 0.90 ☐ 1.20

☐ identified offset within 1 mm of truth

S10 — roll boundary, (38, 38) mm, 3.066 kg

predicted M_{crit} : roll +0.15 / +2.13, pitch -2.41 / +0.12 — **expected to fail**: roll: contact unloaded under the excitation collective

roll ☐ 0.30 ☐ 0.90

pitch ☐ 0.30 ☐ 0.90

☐ the predicted contact lifts at zero applied moment

S11 — pitch boundary, (38, 14) mm, 3.066 kg

predicted M_{crit} : roll -0.57 / +1.41, pitch -2.41 / +0.12

roll ☐ 0.10 ☐ 0.20 ☐ 0.30 ☐ 0.45 ☐ 0.65 ☐ 0.90 ☐ 1.20

pitch ☐ 0.10 ☐ 0.20 ☐ 0.30 ☐ 0.45 ☐ 0.65 ☐ 0.90 ☐ 1.20

☐ identified offset within 1 mm of truth

S12 — pitch boundary, (46, 17) mm, 3.066 kg

predicted M_{crit} : roll -0.48 / +1.50, pitch -2.65 / -0.12 — **expected to fail**: pitch: contact unloaded under the excitation collective

roll ☐ 0.30 ☐ 0.90

pitch ☐ 0.30 ☐ 0.90

☐ the predicted contact lifts at zero applied moment

S13 — loaded, (25, 25) mm, 3.22 kg

predicted M_{crit} : roll -0.25 / +1.83, pitch -2.12 / +0.54

roll ☐ 0.10 ☐ 0.20 ☐ 0.30 ☐ 0.45 ☐ 0.65 ☐ 0.90 ☐ 1.20

pitch ☐ 0.10 ☐ 0.20 ☐ 0.30 ☐ 0.45 ☐ 0.65 ☐ 0.90 ☐ 1.20

☐ identified offset within 1 mm of truth